

Exploratory Data Analysis (EDA)



1000
Total Records



8
Features



0
Missing
Values



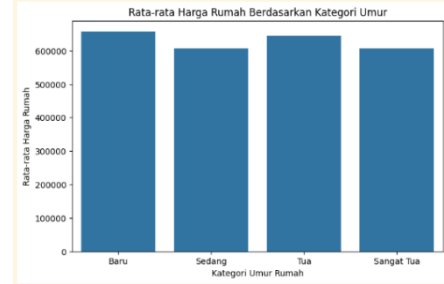
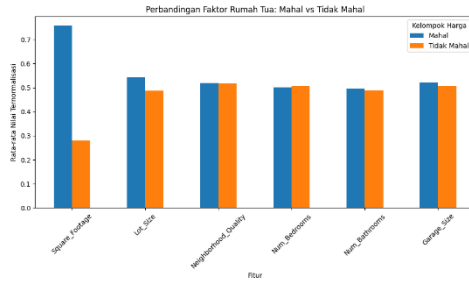
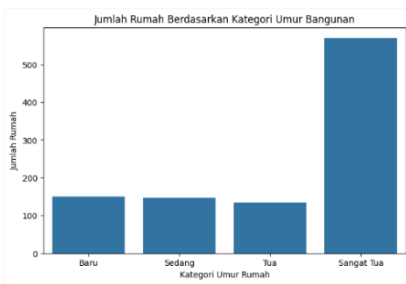
0
Duplicates

DATA QUALITY & BASELINE

	Square_Footage	Num_Bedrooms	Num_Bathrooms	Year_Built	Lot_Size	Garage_Size	Neighborhood_Quality	House_Price
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1.000000e+03
mean	2815.422000	2.990000	1.973000	1986.550000	2.778087	1.022000	5.615000	6.188610e+05
std	1255.514921	1.427564	0.820332	20.632916	1.297903	0.814973	2.887059	2.535681e+05
min	503.000000	1.000000	1.000000	1950.000000	0.506058	0.000000	1.000000	1.116269e+05
25%	1749.500000	2.000000	1.000000	1969.000000	1.665946	0.000000	3.000000	4.016482e+05
50%	2862.500000	3.000000	2.000000	1986.000000	2.809740	1.000000	6.000000	6.282673e+05
75%	3849.500000	4.000000	3.000000	2004.250000	3.923317	2.000000	8.000000	8.271413e+05
max	4999.000000	5.000000	3.000000	2022.000000	4.989303	2.000000	10.000000	1.108237e+06

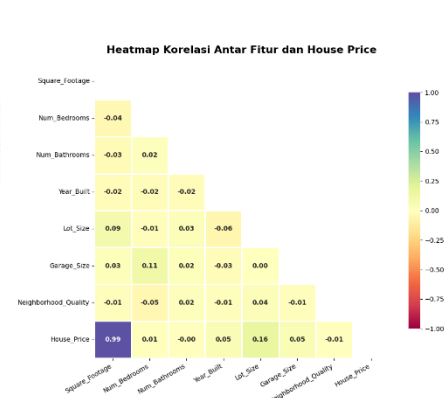
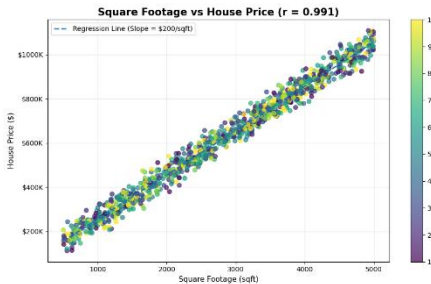
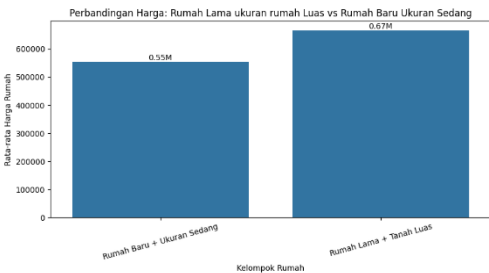
The dataset exhibits sufficient variability across features, providing diverse property characteristics for analysis. The variation observed in the target variable suggests meaningful pricing differences that can support predictive modeling.

Feature Distribution



older properties dominate the dataset distribution, relatively stable average prices across age categories suggest that building age alone provides limited explanatory power. This pattern indicates that pricing behavior is likely driven by interactions among multiple property attributes rather than a single feature.

Key Drivers of House Prices



Larger properties remain more expensive even when compared to newer houses, suggesting that property size has a stronger influence on pricing than building age. This pattern is further supported by strong correlation analysis, highlighting size as the primary driver of price variation.

EXECUTIVE SUMMARY

Exploratory analysis reveals heterogeneous pricing patterns across properties, with building age showing limited influence on price variation. Instead, structural property characteristics appear to contribute more strongly, with property size consistently emerging as the dominant factor associated with house prices.